

No. of Functional Features Included in the Solution

Date	06 JULY 2026
Team ID	SWTID-2026-4324
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Login & Registration	Allows farmers and administrators to securely register and log in.	Authentication	Done	10%
2	Crop Recommendation	Recommends the best crop based on soil type, weather, and season using AI/ML.	Recommendation Engine	Done	20%
3	Weather Forecast Integration	Displays real-time weather updates to help farmers plan agricultural activities.	Weather Module	Done	15%
4	Soil Analysis	Analyzes soil parameters such as pH, moisture, and nutrients for better crop selection.	Soil Analysis Module	Done	15%
5	Fertilizer Recommendation	Suggests suitable fertilizers based on crop and soil condition.	Fertilizer Recommendation	Done	10%
6	Irrigation Management	Recommends optimal irrigation schedules to reduce water wastage.	Irrigation Module	Done	10%
7	Yield Prediction	Predicts expected crop yield using historical and environmental data.	Prediction Engine	In Progress	10%
8	Dashboard & Reports	Provides visual reports, analytics, and farming insights through graphs and dashboards.	Dashboard & Reporting	Done	10%

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	User Authentication, Crop Recommendation, Soil Analysis, Weather Forecast, Fertilizer Recommendation

Additional / Nice-to-Have Features	Irrigation Management, Yield Prediction, Dashboard & Reports
Features Tested & Verified	7 Features Successfully Tested and Verified

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Farmer-friendly interface, Dashboard, Reports	Login Page, Crop Recommendation Screen, Weather Dashboard
2	Backend / Logic	AI-based crop recommendation, Yield prediction, Fertilizer recommendation	Crop Recommendation Engine, Irrigation Logic, Yield Prediction Module
3	Database / Storage	Stores user details, crop data, soil records, weather history	Farmer Database, Crop Database, Soil & Weather Data Storage
4	API / Integration	External API integration for weather and agricultural information	Weather API, Soil Data API, SMS/Email Notification API
5	Security / Authentication	Secure user access and data protection	User Login, Registration, Password Encryption, Role-Based Access